

Exercise 2



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Macroeconomics I

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Analytical Study

a.

What are the steady-state values of physical capital, human capital, and output, all per unit of effective labor?

We have:

- $Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}$
- $\dot{k} = S_K k^\alpha h^\beta - (g + n + \delta)k$
- $\dot{h} = S_H k^\alpha h^\beta - (g + n + \delta)h$

In the fix point, the system has no dynamic, so we have:

$$\dot{k} = 0$$

$$S_K k^\alpha h^\beta - (g + n + \delta)k = 0$$

$$S_K k^\alpha h^\beta = (g + n + \delta)k$$

$$k^{\alpha-1} = (g + n + \delta)h^{-\beta} S_K^{-1}$$

$$k^* = \left(\frac{h^{*\beta} S_K}{g + n + \delta} \right)^{\frac{1}{1-\alpha}} \star$$

$$\dot{h} = 0$$

$$S_H k^\alpha h^\beta - (g + n + \delta)h = 0$$

$$S_H k^\alpha h^\beta = (g + n + \delta)h$$

$$h^{\beta-1} = (g + n + \delta)k^{-\alpha} S_K^{-1}$$

$$h^* = \left(\frac{k^{*\beta} S_K}{g + n + \delta} \right)^{\frac{1}{1-\beta}} \star \star$$

We also have:

$$\begin{aligned}
 S_K k^\alpha h^\beta &= (g + n + \delta)k \\
 S_H k^\alpha h^\beta &= (g + n + \delta)h \\
 \xRightarrow{\text{Divide}} \frac{S_K}{S_H} &= \frac{k}{h} \\
 \implies h^* &= \frac{S_H}{S_K} k^* \quad \star \star \star
 \end{aligned}$$

By substituting $\star \star \star$ in $\star$, we would have:

$$\begin{aligned}
 k^* &= \left(\frac{S_H}{S_K} k^* \right)^\beta S_K \frac{1}{g + n + \delta} \\
 &= k^{*\beta} \left(\frac{S_H^\beta S_K^{1-\beta}}{g + n + \delta} \right)^{\frac{1}{1-\alpha}} \\
 \implies k^* &= \left\{ \left(\frac{S_H^\beta S_K^{1-\beta}}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}} \vee 0 \right\}
 \end{aligned}$$

By substituting $\star \star \star$ in $\star \star$, we would have:

$$\begin{aligned}
 h^* &= \left(\frac{S_K}{S_H} h^* \right)^\alpha S_H \frac{1}{g + n + \delta} \\
 &= \left(\frac{h^{*\alpha} S_K^\alpha S_H^{1-\alpha}}{g + n + \delta} \right)^{\frac{1}{1-\beta}} \\
 \implies h^* &= \left\{ \left(\frac{S_K^\alpha S_H^{1-\alpha}}{g + n + \delta} \right)^{\frac{1}{1-\alpha-\beta}} \vee 0 \right\}
 \end{aligned}$$

b.

What is the growth rate of output per capita in steady state?

We know that the output per effective labor has no Dynamic at the fix point. Output per capita equals to output per effective labor times productivity. So at the fix point, it grows with rate of productivity growth wich is equal to " n ".

c.

Study the effect of changes in s_h and s_k on steady state values of y, k, h .

At fixed point, we have:

$$h = \left(\frac{S_K^\alpha S_H^{1-\alpha}}{g + n + \delta} \right)$$

So by taking derivative, we would have:

$$\begin{aligned} \frac{\partial k}{\partial S_H} &= \frac{(1 - \alpha) \left(\frac{S_H^{1-\alpha} S_K^\alpha}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}}}{S_H(1 - a - b)} \\ \frac{\partial k}{\partial S_K} &= \frac{\alpha \left(\frac{S_H^{1-\alpha} S_K^\alpha}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}}}{S_K(1 - a - b)} \end{aligned}$$

At fixed point, we have:

$$k = \left(\frac{S_H^\beta S_K^{1-\beta}}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}}$$

So by taking derivative, we would have:

$$\begin{aligned} \frac{\partial k}{\partial S_H} &= \frac{\beta \left(\frac{S_H^\beta S_K^{1-\beta}}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}}}{S_H(1 - a - b)} \\ \frac{\partial k}{\partial S_K} &= \frac{(1 - \beta) \left(\frac{S_H^\beta S_K^{1-\beta}}{g + n + \delta} \right)^{\frac{1}{1-\beta-\alpha}}}{S_K(1 - a - b)} \end{aligned}$$

At fixed point, we have:

$$\begin{aligned} y &= \left(\frac{S_H^\beta S_K^{1-\beta}}{g+n+\delta} \right)^{\frac{\alpha}{1-\beta-\alpha}} \left(\frac{S_H^\alpha S_K^{1-\alpha}}{g+n+\delta} \right)^{\frac{\beta}{1-\alpha-\beta}} \\ &= \left(\frac{S_K}{g+n+\delta} \right)^{\frac{\alpha}{1-\beta-\alpha}} \left(\frac{S_H}{g+n+\delta} \right)^{\frac{\beta}{1-\alpha-\beta}} \end{aligned}$$

So by taking derivative, we would have:

$$\begin{aligned} \frac{\partial y}{\partial S_H} &= - \frac{\beta \left(\frac{S_K}{g+n+\delta} \right)^{\frac{\alpha}{1-\beta-\alpha}} \left(\frac{S_H}{g+n+\delta} \right)^{\frac{\beta}{1-\alpha-\beta}}}{S_H(\alpha + \beta - 1)} \\ \frac{\partial y}{\partial S_K} &= - \frac{\alpha \left(\frac{S_K}{g+n+\delta} \right)^{\frac{\alpha}{1-\beta-\alpha}} \left(\frac{S_H}{g+n+\delta} \right)^{\frac{\beta}{1-\alpha-\beta}}}{S_K(\alpha + \beta - 1)} \end{aligned}$$

d.

Linearize the system around the non-trivial fixed point, and find the Eigenvalues associated with it.

Bonus: Do they represent the convergence rate in the proximity of the fixed point? If the answer is negative, try finding the convergence rates.

To linearize the system around the non-trivial fixed point, we need to set up the Jacobian matrix and find the Eigenvalues.

We had the following system of differential equations:

$$\begin{aligned}\dot{k} &= S_K k^\alpha h^\beta - (g + n + \delta)k \\ \dot{h} &= S_H k^\alpha h^\beta - (g + n + \delta)h\end{aligned}$$

The Jacobian matrix at the fixed point (k^*, h^*) is:

$$J = \begin{bmatrix} \frac{\partial \dot{k}}{\partial k} & \frac{\partial \dot{k}}{\partial h} \\ \frac{\partial \dot{h}}{\partial k} & \frac{\partial \dot{h}}{\partial h} \end{bmatrix}_{(k^*, h^*)}$$

Now we should calculate the partial derivatives, to set up the Jacobian matrix:

$$\begin{aligned}\frac{\partial \dot{k}}{\partial k} &= \alpha S_K k^{*\alpha-1} h^{*\beta} - (g + n + \delta) \\ \frac{\partial \dot{k}}{\partial h} &= \beta S_K k^{*\alpha} h^{*\beta-1} \\ \frac{\partial \dot{h}}{\partial k} &= \alpha S_H k^{*\alpha-1} h^{*\beta} \\ \frac{\partial \dot{h}}{\partial h} &= \beta S_H k^{*\alpha} h^{*\beta-1} - (g + n + \delta)\end{aligned}$$

To find the eigenvalues, we must solve the following equation:

$$\det(J - \lambda I) = 0$$

$$\det \begin{bmatrix} \alpha S_K k^{*\alpha-1} h^{*\beta} - (g + n + \delta) - \lambda & \beta S_K k^{*\alpha} h^{*\beta-1} \\ \alpha S_H k^{*\alpha-1} h^{*\beta} & \beta S_H k^{*\alpha} h^{*\beta-1} - (g + n + \delta) - \lambda \end{bmatrix} = 0$$

By calculating the determinant, we will reach to the following equation:

$$-(g + n + \delta + \lambda) \left[\alpha S_K k^{*\alpha-1} h^{*\beta} + \beta S_H k^{*\alpha} h^{*\beta-1} - (g + n + \delta + \lambda)^2 \right] = 0$$

$$(i) \ g + n + \delta + \lambda = 0$$

$$\implies \lambda = -(g + n + \delta) \quad \checkmark$$

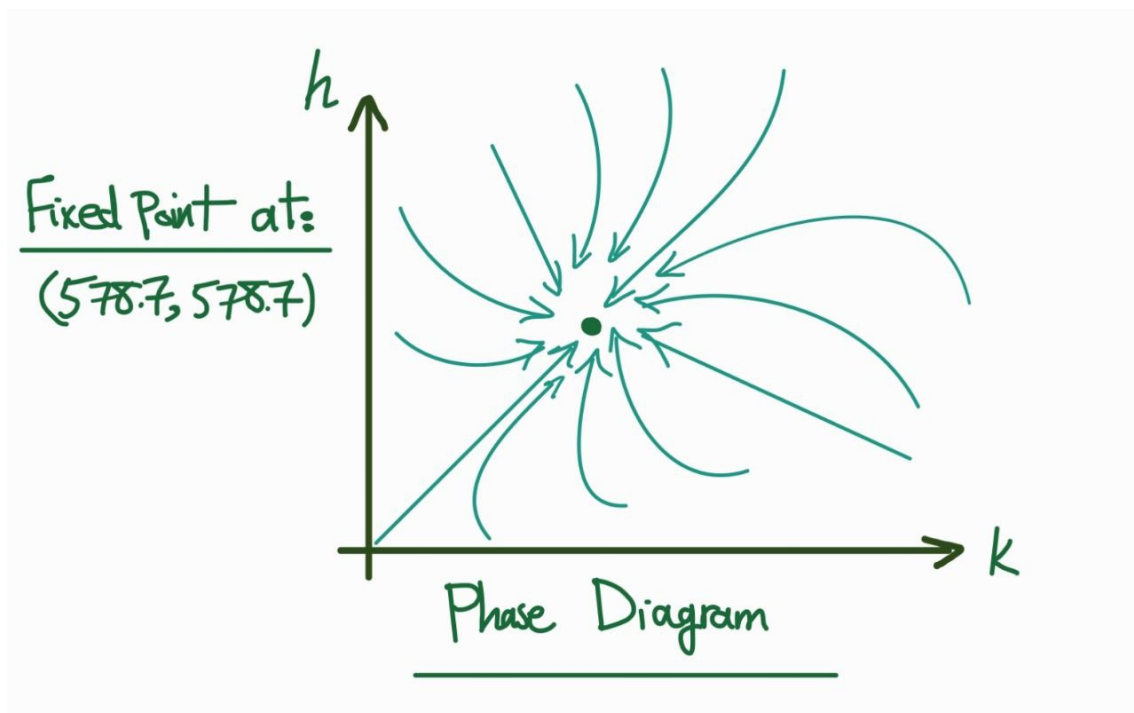
$$(ii) \ \alpha S_K k^{*\alpha-1} h^{*\beta} + \beta S_H k^{*\alpha} h^{*\beta-1} - (g + n + \delta + \lambda)^2 = 0$$

$$\implies \lambda = \sqrt{\alpha S_K k^{*\alpha-1} h^{*\beta} + \beta S_H k^{*\alpha} h^{*\beta-1}} - g - n - \delta \quad \checkmark$$

Quantitative Analysis

e.

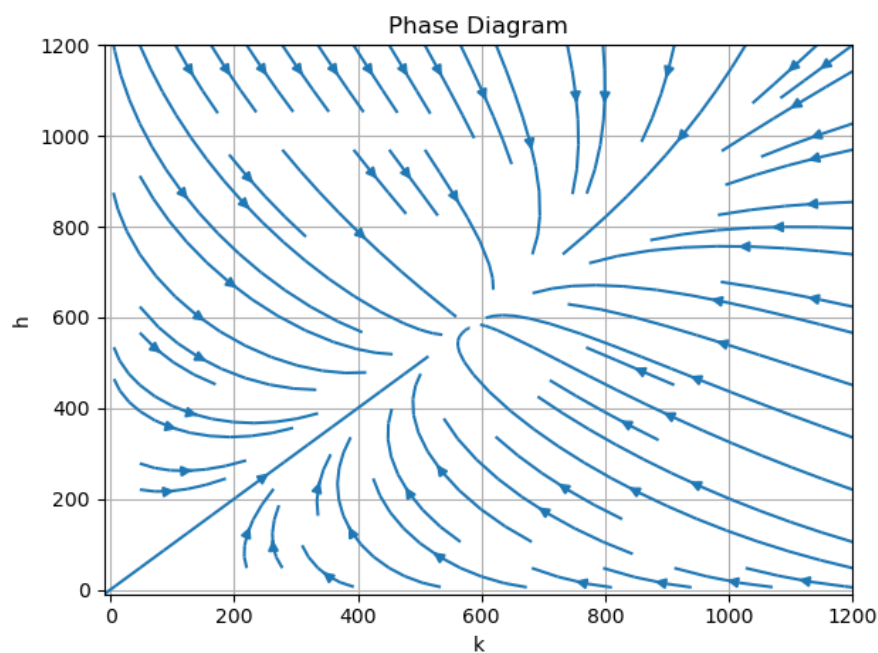
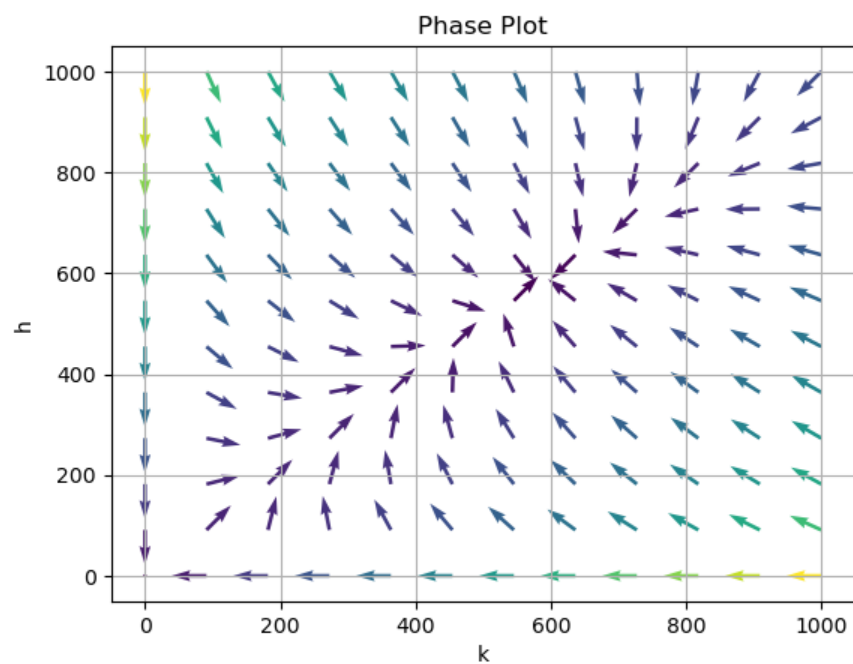
Plot the phase diagram of the system in $h - k$ plane by hand.



f.

Plot the phase diagram of the system in $h - k$ plane using Matlab or Python. (Solve the system using ODE45).

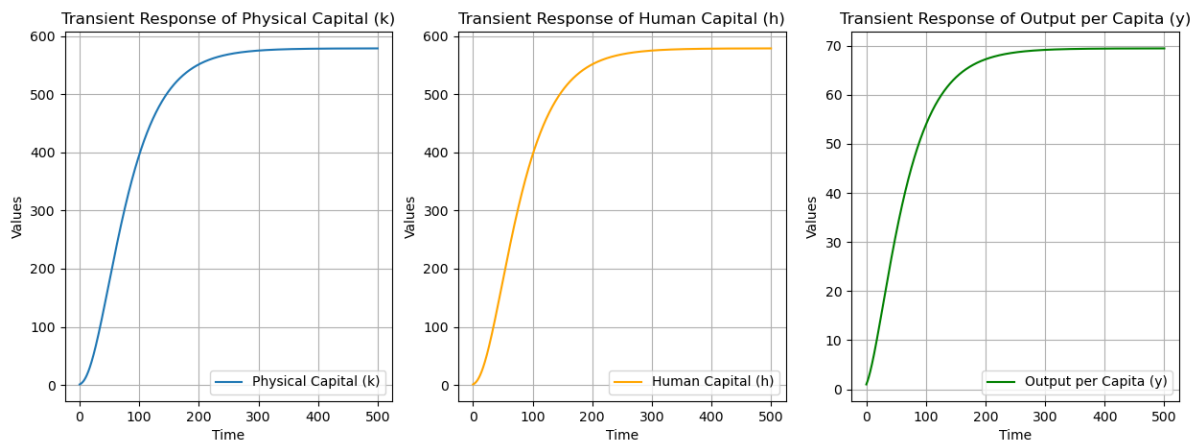
To open the python code for this problem: [Click Here](#)



g.

Choose any arbitrary initial condition, close to non-trivial fixed point of the system, and plot the transient responses for k , h , y .

To open the python code for this problem: [Click Here](#)



h.

Write the equations of motion obtained in part (a) in discrete time. Use “Dynare” to solve the system numerically, and compare your results.

To open the Dynare code for this problem: [Click Here](#)

